9/11/26 Front end Q1A DHANUSH SURYA S 7384

1- Javascript

* JS is a programming language used to create web applications.
* It is opensource and dynamically typed
* We can create dynamic webpages using JS
* It is function oriented language
* JS is single thread architecture

Features:

* Single thread architecture stops execution even if has single error
* API Access
* dynamicall typed
* Interpreted language.

2. var, let and const

*Scope is used to define where a declared variable can be used or called

* Redeclaration is the process of declare a variable with a name of existing variable
* Reassign is the process of assigning a value to a variable that the variable may have a value already or not

var:

*var is a type of variable declare where can do all 3 above process

*var is function scope or global scope. It cannot be bounded by a block scope

let:

* Let is a type of variable declaration introduced in ES6.

*It is block scope if inside a function or a block or a conditional statement

*redeclaration cannot be done but we can reassign a value.

const.

* const is a type of declaration introduced in ES

*It is also a block scope

* we cannot reassign or redeclare const variable.

3 Datatypes in JS

* Datatypes are type of data that stored in a variable

* Types of data-types
  * primitive
  * non-primitive

Primitive

* String, Number, Boolean and null, undefined

```
let string = "React"
let num    = 4
let Bool   = true
let nul    = null
let undef  = //undefined
```

Non-primitive,

* It includes array, object, array of objects

* Array is collection of element
* Object is matching value with a key
* Array of object is collection object

```
let arr = [1,2,3,3,4,5,6]
let obj = {name:"value", num:4}
```

let arrobj = [{obj}, {obj}, {obj}]

4. Function

*It is a block of code we can use it again that can be called after declaration.

* codes needs to be written inside curly braces{}

* Here we can declare data inside function or can be passed as a argument

* Arguments can be a value, variable or another function.

* Return Statement should be inside function

Normal Function: function count () {
let a = 1
let count = a+1
return count
}

Expression => let count = function () {
}

Arrow function

const caunt = () => {
return
}

5) == or ===

== is a comparative operator
(loose equal)
used to compare whether both values are same or not

Ex:
num = 4
num == 4 //true
num == 10 //false
num == "4" //true checks String to o

=== Also a comparative operator
Strict equal

Ex: num = 4
num === 4 //true
num === "4" false

Cannot accept String vs. Number

6 React

* React is a JS Library used to build web applications.
* It use component based architecture
* we have React 3 and vite
* previous version use Eslint current version use Oxlint
* It uses v-DOM instead of DOM

* Virtual DOM is like version of DOM

* Using V-DOM we can update re-renders a specific component instead of whole webpage

7. components:

* It is a reusable code used by react to design the webpage

* Each webpage consist of a number of components

* components are created using the combination of both html and JS

* Each component contain a default function and addition function

React use component

* It has a property called as rendering

* React renders the component automatically without refreshing the page.

8. JSX → JS XML → JS External Markup Language.

* we used HTML inside Javascript code

* It cannot be directly compiled, It is first transfred to JS file using CRLF then sent to compiler

9. Props and state

Props	State
* Data transfer between component	* It is a data management unit
* It provides read only data to its child components for reading and displaying	* It stores data dynamically and update will be immediately visible on the webpage
* props can only be sent to child component and updated in parent component	* It only works on class at initial, now to functionality is transferred to functional component using ~~web~~ use state ()..

10 Virtual Dom.

* V-Dom is liter version of Dom
* Identifies components and re-render it without reloading webpage
* It uses Reconciliation method to identify changes
* V-Dom compares to it previous version.
* V-Dom update state automatically when State change